

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Civil Engineering End Semester Examination Academic Year: 2025-26 (Semester – I)		
Class: Final Year B.Tech. (Civil Eng.)	Subject: Structural Dynamics.		Sub Code: PEC– CE410
Date: 29/11/25	Time: 13:30 to 15:30		Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Nonprogrammable calculator is permitted.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Question	Marks	CO	BT
Q 1 a	Explain the difference between static and dynamic loads with examples.	05	CO1	BT1
b	Derive the equation of motion for a damped single degree of freedom system subjected to a harmonic force.	05	CO3	BT3
Q 2 a	Calculate the natural time period of structure for a modal analysis refers the following data: $M_1 = 1.6 \text{ kg}$, $M_2 = 1.25 \text{ kg}$, $M_3 = 1 \text{ kg}$ $K_1 = 2.304 \times 10^4 \text{ N/mm}$, $K_2 = 5.388 \times 10^4 \text{ N/mm}$, $K_3 = 1.5 \times 10^4 \text{ N/mm}$ Using characteristics equation.	10	CO4	BT4
Q 3 a	Explain the concept of mode shapes and orthogonality principle in MDOF systems.	05	CO4	BT4
b	A hospital building in Zone V ($Z = 0.36$) has seismic weight $W = 20,000 \text{ kN}$, $I = 1.5$, $R = 4$, and $S_a/g = 2.5$. Find the design base shear.	05	CO4	BT4
Q 4 a	Explain Vanello Stodola and matrix iteration method.	05	CO2	BT2
b	Using 3 masses $M_1 = M_2 = M_3 = 2 \text{ kg}$ and 4 springs $K_1 = K_2 = K_3 = 60 \text{ N/m}$ Calculate the natural freequency if the mode shape $[1:1:1]$	05	CO5	BT4

Q 5	a	Derive the Dynamic load factor for a SDOF under an impulse load for rectangular section using Duhamel's integral.	05	CO4	BT5
	b	A system has natural frequency $\omega_n = 20$ rad/sec and damping ratio $\zeta = 0.05$ find a) Damped natural frequency ω_d b) Find δ c) Time for amplitude to reduce half.	05	CO4	BT4
Q 6	a	Derive the expression for fundamental frequency of a MDOF system by using Rayleigh's ritz method.	05	CO5	BT6
	b	State the advantages and limitations of using response spectrum analysis.	05	CO5	BT4

== ALL THE BEST ==

Course Outcomes:

CO 1	To recall fundamentals of Mathematics and Engineering Mechanics.
CO 2	To Understand the effects of system/model parameters on dynamic response.
CO 3	To apply and formulate dynamic equilibrium equations for SDOF and MDOF system
CO 4	To analyze SDOF and MDOF systems using classical and numerical methods.
CO 5	To evaluate and obtain the response of SDOF, MDOF systems and conduct modal analysis of MDOF Systems.

Bloom's Taxonomy (BT)

BT1-Remember, BT2- Understand, BT3- Apply, BT4- Analyse, BT5- Evaluate, BT6-Create



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
End Term Examination (Semester - I)
Academic Year: 2025-26

Class: Final Year-B.Tech.

Subject: Elective: Construction Engineering & Management

Subject Code: PEC-CE408

Date: 29/11/2025

Time: 13:30 pm to 15:30 pm

Max. Marks: 60

Note:

- i. Non-programmable calculator is permitted.
- ii. Figures to right indicate full marks.

Q.No.		Question	Marks	CO	BT																								
Q1	a	Explain in detail the various modern techniques used in Project Management.(any 3)	06	CO1	BT1																								
	b	The following activities form part of a construction project. Draw network, find critical path, and compute total project duration. <table><tr><th>Activity</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th></tr><tr><td>Predecessor</td><td>—</td><td>A</td><td>A</td><td>B</td><td>B,C</td><td>D</td><td>E,F</td></tr><tr><td>Duration (days)</td><td>4</td><td>6</td><td>5</td><td>7</td><td>3</td><td>8</td><td>6</td></tr></table>	Activity	A	B	C	D	E	F	G	Predecessor	—	A	A	B	B,C	D	E,F	Duration (days)	4	6	5	7	3	8	6	05	CO1	BT3
Activity	A	B	C	D	E	F	G																						
Predecessor	—	A	A	B	B,C	D	E,F																						
Duration (days)	4	6	5	7	3	8	6																						
Q2	a	Define and differentiate Optimistic, Pessimistic, Most-likely, and Expected time. Give one example for each.	04	CO2	BT2																								
	b	A PERT activity has the following time estimates: to = 5 days, tm = 11 days, and tp = 17 days Calculate: i) Expected duration ii) Variance iii) Standard deviation iv) Probability of completing the task within 10 days and 12 days (use Z-score)	05	CO2	BT3																								
Q3	a	Explain in detail any three construction methods from the following: 1) Slip-form construction 2) Prefabrication 3) Dewatering techniques 4) Underpinning methods	06	CO3	BT2																								
	b	A contractor needs to provide 45 m³/h of concrete. a) Mixer capacity = 0.6 m³ b) Mixing time = 12 minutes per batch Determine: i) Number of mixers required ii) Productivity improvement if batch time reduces to 8 minutes	06	CO3	BT3																								
Q4	a	Explain the factors affecting selection of construction equipment. A bulldozer has: 1) Machine speed = 4 km/h 2) Blade capacity = 2.4 m³ 3) Job efficiency = 80% 4) Load factor = 0.85	04	CO3	BT2																								
	b		06	CO3	BT3																								

		5) Cycle time = 1.2 minutes Calculate: i) Output per hour ii) Number of bulldozers required for 600 m ³ /day (8 marks)																							
Q5	a	Explain Resource Levelling and Resource Smoothing with neat illustrations.	04	CO3	BT2																				
	b	Explain the concept of crashing of a project. Write and explain criteria for selection of an activity for crashing.	04	CO3	BT2																				
Q6	a	Explain the following with examples: i) Item-rate contract ii) Cost-plus contract	04	CO4	BT2																				
	b	A company is considering purchase of a new machine. Two alternative machines (X and Y) have been suggested, each having an initial cost of Rs. 20,00,000 and requiring Rs. 1,00,000 as additional working capital at the end of 1st year. Earnings after taxation are expected to be as follows: <table><tr><th rowspan="2">Year</th><th colspan="2">Cash flows (Rs.)</th></tr><tr><th>Machine X</th><th>Machine Y</th></tr><tr><td>1</td><td>200000</td><td>600000</td></tr><tr><td>2</td><td>600000</td><td>800000</td></tr><tr><td>3</td><td>800000</td><td>1000000</td></tr><tr><td>4</td><td>1200000</td><td>600000</td></tr><tr><td>5</td><td>800000</td><td>400000</td></tr></table> The company has target of return of capital of 10% and on this basis, you are required to prepare the profitability of the machines and state which alternative you consider financially preferable using NPV method. Consider present value for Rs.1/- at 10% cut-off- rate.	Year	Cash flows (Rs.)		Machine X	Machine Y	1	200000	600000	2	600000	800000	3	800000	1000000	4	1200000	600000	5	800000	400000	06	CO4	BT4
Year	Cash flows (Rs.)																								
	Machine X	Machine Y																							
1	200000	600000																							
2	600000	800000																							
3	800000	1000000																							
4	1200000	600000																							
5	800000	400000																							

== ALL THE BEST ==

NORMAL DISTRIBUTION TABLE

Area Under Standard Normal Distribution

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO	On successful completion of this course, students should be able to:
CO1	Define the fundamental principles of management theories
CO2	Understand the feasibility analysis in construction management and network analysis tools for cost and time estimation
CO3	Apply theoretical and practical aspects of construction management techniques to achieve project goals
CO4	Analyze contemporary construction management tools and methodologies in Indian context.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
Academic Year: 2025-26 (semester I)
End Semester Examination

Class: Final Year B. Tech.

Course Name: Elective: Water
Conservation & Management

Course Code: PEC-CE407

Date: 18.11.2025

Time: 13:30 to 15:30

Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Explain the model for estimating the soil erosion loss (USLE).	5	CO2	BT2
	b	Describe any three structures of soil erosion control.	3	CO2	BT2
Q.2	a	Write the criteria for selection of site for farm pond and percolation tank	4	CO1	BT2
	b	Illustrate the need of soil and water conservation	3	CO4	BT2
Q.3	a	Explain the mathematical modelling for operation of reservoir.	4	CO3	BT2
	b	Calculate the peak runoff rate for a 10 years return period from a micro-watershed area of 75 hectares. The watershed is divided into three parts, based on its land use and soil texture, in which first part of 25 hectares with 1% slope is under cultivation, second part of 30 hectares with a slope of 7% is under pasture and rest of the land with slope of 12% is under farm forestry. The maximum length of flow path is 3000 meters to the outlet. The average slope of channel is 5% and the maximum rainfall intensity of 1-h rainfall is 5 cm/h during the 10 year return period.	4	CO3	BT3
Q.4	a	Describe the areal aspect of drainage basin.	4	CO3	BT2
	b	Explain the objectives of watershed management	3	CO2	BT2
Q.5	a	Describe the participative irrigation management (PIM)	6	CO3	BT2
	b	What is cropping pattern? Explain each type of cropping pattern detail.	6	CO1	BT2
	c	Explain the provisions in PIM Acts.	3	CO3	BT2
Q.6	a	What is land levelling? Explain the criteria for land levelling	7	CO4	BT2
	b	Describe the considerations in land grading design	8	CO6	BT3

==== ALL THE BEST ====

Bloom's Taxonomy (BT)

BT1 - Remember. BT2 - Understand. BT3 - Apply. BT4 - Analyze. BT5 - Evaluate. BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Recall the concepts of WRE-I & II
CO 2	Understand the concepts related to Soil and water conservation techniques and watershed management
CO 3	Apply the knowledge of theories and equations to drainage design, irrigation system management, etc.
CO 4	Analyse the different practice for soil and water conservation.

Table 2.1. Values of 'C' for use in Rational formula

Sl. No.	Land use and Topography	Soil Type		
		Sandy loam	Clay and silt loam	Tight clay
1.	Cultivated land			
	(a) Flat	0.30	0.50	0.60
	(b) Rolling	0.40	0.60	0.70
	(c) Hilling	0.52	0.72	0.82
2.	Pasture land			
	(a) Flat	0.10	0.30	0.40
	(b) Rolling	0.16	0.36	0.55
	(c) Hilling	0.22	0.42	0.60
3.	Forest land			
	(a) Flat	0.10	0.30	0.40
	(b) Rolling	0.25	0.35	0.60
	(c) Hilling	0.30	0.50	0.60
4.	Populated area			
	(a) Flat	0.40	0.55	0.65
	(b) Rolling	0.50	0.65	0.80

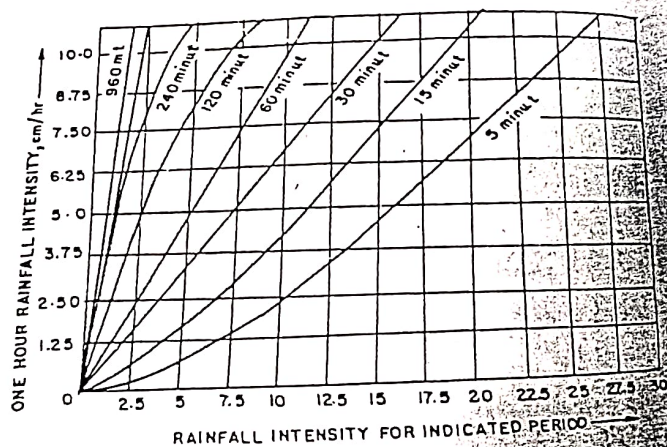


Fig. 2.3. Relationship between intensities of indicated duration and 1-hr rainfall intensity



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
End Semester Examination
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Course: Advanced Structural Analysis

Course Code: PEC-CE404

Date: 18.11.2025

Time: 01:30 pm to 03:30 pm

Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Non-programmable calculators are allowed.

- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Solve either question 1 or question 2 (Any one)

Q.N.	Statement	Marks	CO	BT
Q.1	For the frame shown in Figure 1 (a), use the stiffness method to; i) Determine the deflection and rotation at B, ii) Determine all the reactions at support, and iii) Draw Shear force and Bending moment diagrams Take $E = 200 \text{ GPa}$, and $I = 60 \times 10^6 \text{ mm}^4$, $A = 600 \text{ mm}^2$, refer the node numbering, member numbering, and degrees of freedom as shown in the figure 1 (b)	30	CO 2, 4	BT 4

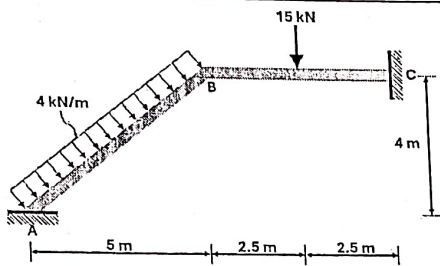


Figure 1 (a)

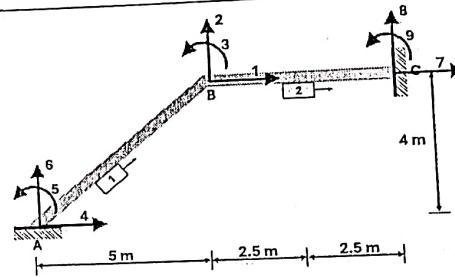


Figure 1 (b)

OR

Q.2	For the truss shown, use the stiffness method to: i) Determine the end forces of each member. ii) Determine the deflections of the loaded joint. Take $E = 200 \text{ GPa}$, $A = 1000 \text{ mm}^2$.	30	CO 2, 4	BT 4
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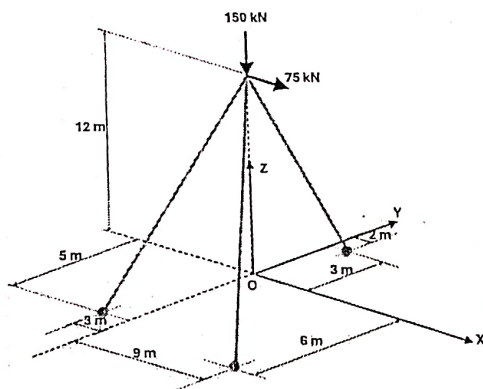


Figure 2 (a)

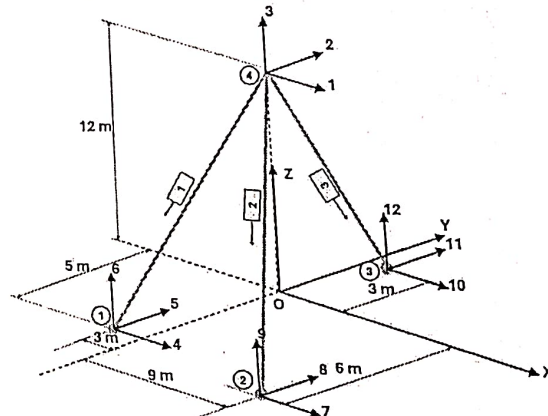


Figure 2 (b)

Q 3) a)	Explain, with a suitable example, how to apply an effective node numbering scheme to minimize the bandwidth of a matrix.	10	CO 3	BT 3
Q 3) b)	Explain 3 coordinate systems generally used in the finite element analysis with example.	06	CO 4	BT 2
Q 4) a)	Enlist and explain the types of nodes in a Finite Element Analysis.	08	CO 4	BT 2
Q 4) b)	Explain following in the context of finite element analysis. i) Discretization ii) Aspect ratio	06	CO 4	BT 2

=== ALL THE BEST ===



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
End Semester Examination
Academic Year: 2025-26 (Semester I)

Class: Final Year B.Tech

Course: Foundation Engineering

Course Code: PCC-CE403

Date: 22.11.2025

Time: 13:30 to 15:30

Max. Marks: 60

Note: Mobile phones and programmable calculators are strictly prohibited.

Q.N.	Questions	Marks	CO	BT
Q.1	Explain plate load test in details with neat sketch.	06	1	2
Q.2	Determine the ultimate, net ultimate, net safe, and safe bearing capacities of a 4 m square footing placed 1.8 m deep using Terzaghi's theory. The soil has $\gamma = 20 \text{ kN/m}^3$, $C = 18 \text{ kN/m}^2$, and $\phi = 20^\circ$. Assume local shear failure with $N_c = 11.8$, $N_q = 3.8$, and $N_\gamma = 1.3$, and use a factor of safety of 3.	06	2	3
Q.3	A footing 3 m x 1.5 m in plan transmits a pressure of 160 kN/m^2 on a cohesive soil having $E = 15 \times 10^3 \text{ kN/m}^2$ and $\mu = 0.25$. Determine the immediate settlement in mm at the centre, assuming the footing to be (a) flexible and (b) rigid (refer Table no.1).	06	3	3
Q.4	Explain vibroflotation method of ground improvement technique in detail with neat sketch.	06	2	2
Q.5	Define, with neat sketches, strip footing, strap footing, and combined footing, and also describe the various types of combined footings.	06	5	2
Q.6	Explain under-reamed piles in detail with a neat sketch.	06	3	2
Q.7	A group of 16 piles arranged in a square pattern with diameter and length of pile as 300 mm and 9 m respectively, is used as foundation in soft clay deposit. Taking unconfined compressive strength of clay as 140 kN/m^2 and the pile spacing as 1 m c/c, find the load carrying capacity of the pile individually and as a group. Take adhesion factor as 0.70.	06	4	3
Q.8	Find the allowable pile load using the Engineering News formula, for a given 20 kN drop hammer, a 2.5 m drop height, and a total penetration of 60 mm in the last five blows of a single-acting steam hammer. Also determine the allowable load for the same conditions using a double-acting steam hammer with a 100 cm^2 piston area and a downstroke steam pressure of 175 kN/m^2 .	06	4	3

Q.9	A retaining wall 16 m high retains sand with $\phi = 30^\circ$ and unit weight 18 kN/m^3 upto a depth of 8 m from top. From 8 m to 16 m, the material is a cohesive soil with cohesion of 30 kN/m^2 and $\phi = 20^\circ$. Unit weight of cohesive soil is 20 kN/m^3 . A uniform surcharge of 100 kN/m^2 acts on the top of soil. Determine the active lateral earth pressure at the ground surface, at depths of 8 m and 16 m. Also, draw the earth-pressure distribution diagram and calculate the total active thrust acting on the retaining wall.	12	5	3
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Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Recall the fundamentals of building construction and geotechnical engineering.
CO 2	Identifying the applicability of different foundation design applicable to various types of soil.
CO 3	Apply the different design philosophy of foundation applicable to soil.
CO 4	Analysis various types of shallow and deep foundations.
CO 5	Create understanding applicability of foundation design for different types of soil.

Table no.1

Shape	I _f for Flexible Foundation			I _f for Rigid Foundation
	Centre	Corner	Average	
Circle	1.0	0.64	0.85	0.86
Square	1.12	0.56	0.95	0.82
Rectangle				
L/B= 1.5	1.36	0.68	1.2	1.06
L/B= 2	1.52	0.76	1.3	1.2
L/B= 5	2.10	1.05	1.83	1.70
L/B= 10	2.52	1.26	2.25	2.10
L/B= 100	3.38	1.69	2.96	3.40